Positive-Velocity Goldfish Flow: Global Root Interlacing and One-Carrier Scattering

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Abstract

For arbitrary ordered real positions and strictly positive velocities, we prove that the Calogero goldfish polynomial pencil has simple real roots at every real time. A decreasing partial-fraction map gives exact signed-time interlacing, global collision avoidance, and strict coordinate monotonicity. The fixed roots of the invariant polynomial become scattering anchors: we derive their positive first inverse-time coefficients and, by an exact Vieta law, the intercept and first correction of the unique ballistic root. The carrier moves from the leftmost incoming rank to the rightmost outgoing rank, with sharp failures on the boundary of the positive cone.

1 Polynomial linearization and global interlacing

Fix $N \geq 2$, positions $x_1 < \cdots < x_N$, and velocities $v_i > 0$. Define

$$P(z) = \prod_{i=1}^{N}(z - x_i), \qquad Q(z) = \sum_{i=1}^{N} v_i \prod_{j \neq i}(z - x_j), \qquad p(z,t) = P(z) - tQ(z). \tag{1}$$

Let $y_1 < \cdots < y_{N-1}$ denote the roots of Q.

Theorem 1 (Complete positive-cone flow). *For every $t \in \mathbb{R}$, $p(\cdot, t)$ has N simple real roots $z_1(t) < \cdots < z_N(t)$. They obey*

$$t > 0: \quad x_i < z_i(t) < y_i \ (i < N), \qquad x_N < z_N(t), \tag{2}$$

$$t < 0: \quad z_1(t) < x_1, \qquad y_i < z_{i+1}(t) < x_{i+1} \ (i < N). \tag{3}$$

Every $\dot z_i$ is positive, $\sum_i \dot z_i = V := \sum_i v_i$, and

$$\ddot z_i = 2 \sum_{j \neq i} \frac{\dot z_i \dot z_j}{z_i - z_j}. \tag{4}$$

Thus (1) is a complete collision-free goldfish solution with $z_i(0) = x_i$ and $\dot z_i(0) = v_i$.

Proof. Off the poles x_i, partial fractions give

$$R(z) := \frac{Q(z)}{P(z)} = \sum_{i=1}^{N} \frac{v_i}{z - x_i}, \qquad R'(z) = -\sum_{i=1}^{N} \frac{v_i}{(z - x_i)^2} < 0. \tag{5}$$

Hence $R = 0$ has exactly one root y_i in each interval (x_i, x_{i+1}). When $t \neq 0$, $p(z,t) = 0$ is equivalent to $R(z) = 1/t$. Strict decrease and the one-sided pole limits give exactly the intervals in (2)–(3), including the sign-selected exterior interval. All roots are simple. At $t = 0$ they are the simple roots of P, so this construction covers all real time.

For $t \neq 0$, differentiation of $R(z_i(t)) = 1/t$ gives

$$\dot z_i = -\frac{1}{t^2 R'(z_i)} > 0. \tag{6}$$

At zero, implicit differentiation instead gives $\dot z_i(0) = Q(x_i)/P'(x_i) = v_i$. The coefficient of z^{N-1} in p gives the exact law

$$\sum_i z_i(t) = \sum_i x_i + Vt, \tag{7}$$

and therefore $\sum_i \dot z_i = V$. Differentiating $p(z,t)=\prod_i(z-z_i(t))$ in t shows that the time-independent Q is

$$Q(z)=\sum_i \dot z_i(t)\prod_{j\neq i}(z-z_j(t)). \tag{8}$$

At a root z_i, elementary logarithmic differentiation gives

$$\frac{p_{zz}}{p_z}=2\sum_{j\neq i}\frac{1}{z_i-z_j}, \tag{9}$$

$$\frac{Q'}{p_z}=\sum_{j\neq i}\frac{\dot z_i+\dot z_j}{z_i-z_j}. \tag{10}$$

Twice differentiating $p(z_i(t),t)=0$ and inserting (9)–(10) yields (4). Finally, $p(z,t+s)=p(z,t)-sQ(z)$ proves the exact flow group law. □

2 One-carrier scattering and boundary sharpness

Put

$$M=\sum_i v_i x_i, \qquad c=\frac{M}{V}, \qquad \beta_i=-\frac{P(y_i)}{Q'(y_i)}, \qquad B=\sum_{i=1}^{N-1}\beta_i. \tag{11}$$

Theorem 2 (Two-ended rank-transfer scattering)**.** *Every β_i is positive. As $t\to-\infty$,*

$$z_1(t)=Vt+c+\frac{B}{t}+O(t^{-2}), \tag{12}$$

$$z_{i+1}(t)=y_i-\frac{\beta_i}{t}+O(t^{-2}) \quad (i<N), \tag{13}$$

whereas, as $t\to+\infty$,

$$z_i(t)=y_i-\frac{\beta_i}{t}+O(t^{-2}) \quad (i<N), \tag{14}$$

$$z_N(t)=Vt+c+\frac{B}{t}+O(t^{-2}). \tag{15}$$

All expansions differentiate termwise. Thus the incoming velocity vector tends to $(V,0,\dots,0)$ and the outgoing vector to $(0,\dots,0,V)$ in rank order.

Proof. At a simple zero y_i of Q, the implicit-function theorem with $s=1/t$ applied to $sP(z)-Q(z)=0$ gives

$$z=y_i+\frac{P(y_i)}{tQ'(y_i)}+O(t^{-2})=y_i-\frac{\beta_i}{t}+O(t^{-2}). \tag{16}$$

Since $y_i\in(x_i,x_{i+1})$, the signs of $P(y_i)$ and $Q'(y_i)$ are opposite; hence $\beta_i>0$. Comparing the two leading coefficients in $Q=V\prod_i(z-y_i)$ yields

$$\sum_{i=1}^{N-1} y_i=\sum_{i=1}^{N} x_i-c. \tag{17}$$

Subtracting all finite expansions (16) from the exact sum (7), and using (17), gives the ballistic expansions including the B/t term. Interlacing (2)–(3) fixes their ordered ranks. Analyticity in s justifies differentiated remainders and completes the velocity claim. □

The strict positive cone is sharp. For $x=(0,1)$ and $v=(1,0)$, $p=(z-1)(z-t)$ collides at $t=1$. For $v=(1,-1)$, $p=z^2-z+t$ has a double root at $t=1/4$ and nonreal roots afterwards. The $N=1$ face is a free particle; coincident initial positions are singular; all-negative velocities follow by time reversal but are not merged into Theorems 1–2.

Source lineage

References

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